

CHAPTER TWO

GEOLOGY

2.1 General Geology

Cox's Bazar District in southeastern Bangladesh exhibits a complex geological history influenced by tectonic processes and sedimentary deposition. The region is situated within the Bengal Basin, which extends into parts of eastern India (West Bengal). The formation of the Bengal Basin can be traced back to the late Mesozoic period, approximately 125 million years ago, during the fragmentation of the Gondwana supercontinent. This fragmentation set the stage for the initiation of the basin.

Around 55-50 million years ago, the collision between the Indian and Eurasian Plates occurred, leading to the formation of the Himalayas. The Himalayan range serves as a major source of sediment for the Bay of Bengal. The tectonic collision between the Indian and Burmese plates during the Eocene-Oligocene period resulted in the creation of the Indo-Burmese ranges. The folding of the western part of these ranges occurred within the past few million years.

Sediment deposition within the Bengal Basin has been strongly influenced by deltaic processes since the Oligocene epoch. The palaeo-Brahmaputra river, flowing from the east, and the palaeo-Ganges river system, originating from the west, have transported sediments into the region. These variable fluvial systems have contributed to the deposition of sediments on the late Miocene shelf. The sedimentary processes observed in this area resemble those seen in the modern-day shelf, characterized by high sediment loads and seasonal fluctuations in discharge.

In Cox's Bazar District, the study areas are characterized by two major sedimentary formations: the early to middle Miocene Bhuban Formation and the middle to late Miocene BokaBil Formation. These formations consist of thick accumulations of sandstones and mudstones that were deposited during repeated cycles of transgression and regression. The sediments have been sourced from neighboring orogenic belts.

Deeper parts of the basin in Cox's Bazar District feature the Neo-geneSurma Group sediments, which can reach a thickness of around 4 kilometers in the eastern fold belts. The lithology of the

region primarily consists of fine to medium-grained sandstones, silty shale, shale, sandy siltstone, siltstone, conglomerates, and other sedimentary rock types.

The geological sequence in Cox's Bazar District showcases the interplay between tectonic activity, fluvial systems, and sedimentary deposition. It has resulted in the formation of diverse sedimentary rocks, landforms, and the iconic sandy beach that stretches along the Bay of Bengal. The ongoing processes of erosion, deposition, and tectonic activity continue to shape the geological features of the district.

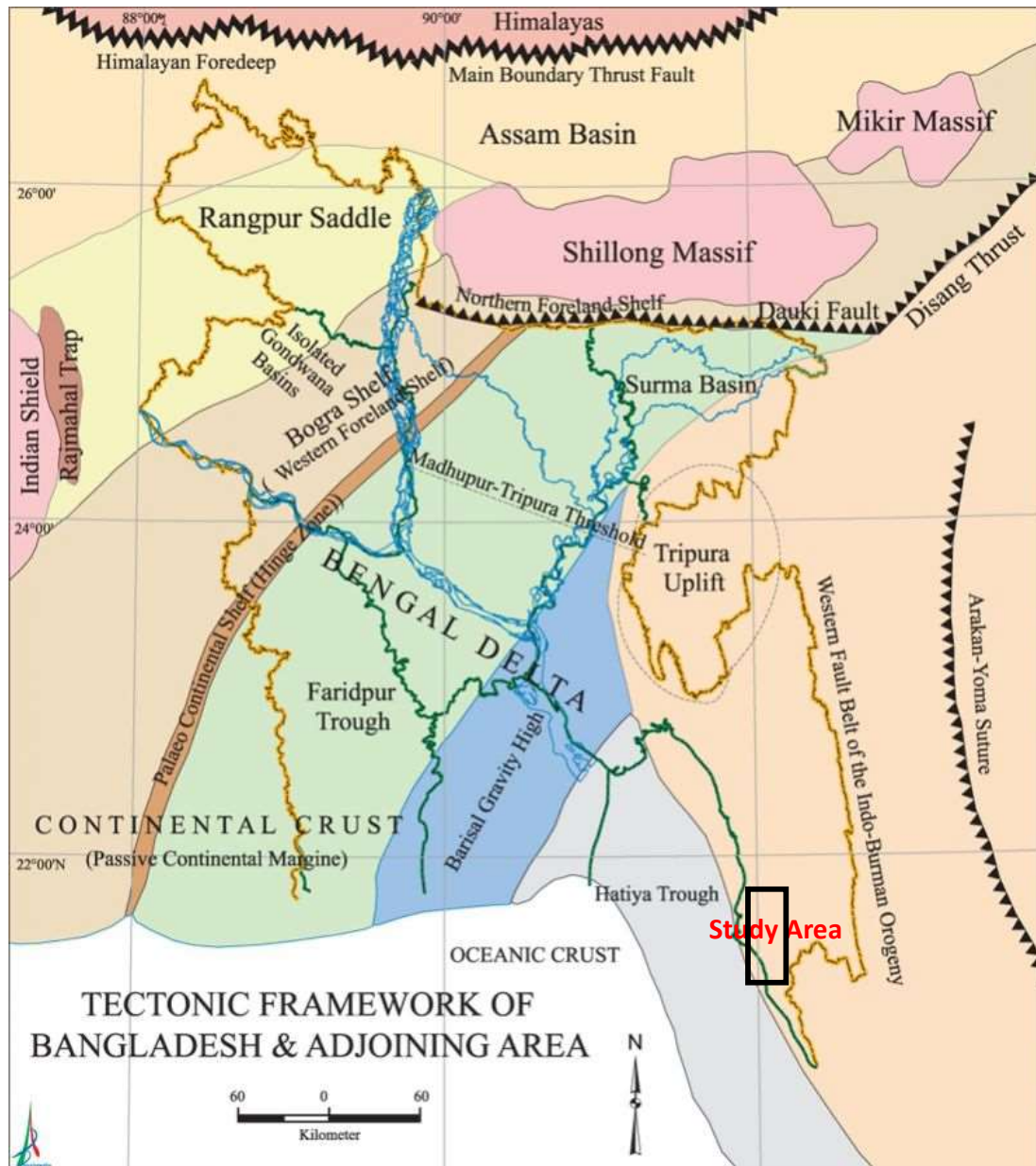


Figure 2.1: Tectonic setup map of Bangladesh. (Guha, 1978 ; GSB, 1990).

2.2 Regional Tectonic Setting

The Bengal Basin is a geological feature situated between the Indian Shield platform to the west and the Indo-Burman folded belt to the east. It is bounded by the Shillong Plateau in the north and extends into the Bay of Bengal in the south, with the current shelf edge located around the 200-meter depth contour. The basin is believed to have formed primarily over a remaining portion of an ancient ocean basin and a rifted continental margin in eastern India. It is known that at least the northern and northwestern parts of the basin are underlain by continental crust, as indicated by Khan and Agarwal (1993) and the Board of Geology and Mining Corporation (BOGMC, 2000). Conversely, the southern and southeastern parts of the basin are underlain by oceanic crust.

Taking a regional perspective, the tectonic features surrounding the Indian subcontinent are believed to have initiated with the northward movement of the Indian plate since the Cretaceous period, culminating in its collision with the Tibetan landmass in the early to middle Eocene epoch. Multiple independent geological and geophysical pieces of evidence support the notion that during a significant portion of the Tertiary period, the Indian plate was subducting beneath both the Tibetan and Burma plates (Curry et al., 1982; Dasgupta and Nandy, 1995). To the north, the collision between the Indian and Tibetan plates resulted in the formation of the east-west-trending Himalayas, while to the east, the subduction process gave rise to the roughly north-south-trending Indo-Burman folded belt, with the eastern folded belt (EFB) being an extension of it towards the west (Lohmann, 1995). The Indo-Burman Ranges and the Andaman Islands constitute a continuous belt that exhibits various geological characteristics of an arc trench system (Dasgupta and Nandy, 1995). The development of the EFB is closely linked to the eastward subduction of the Indian plate within this arc-trench setting (Gani and Alam, 1999). Numerous authors have studied the tectonic framework and structural setting of the Bengal Basin. Notable accounts include the works of Murphy and Staff from BOGMC in 1988, Lohmann in 1995, Alam in 1997, and Sikder in 1998. These studies provide valuable insights into the geological characteristics of the basin.

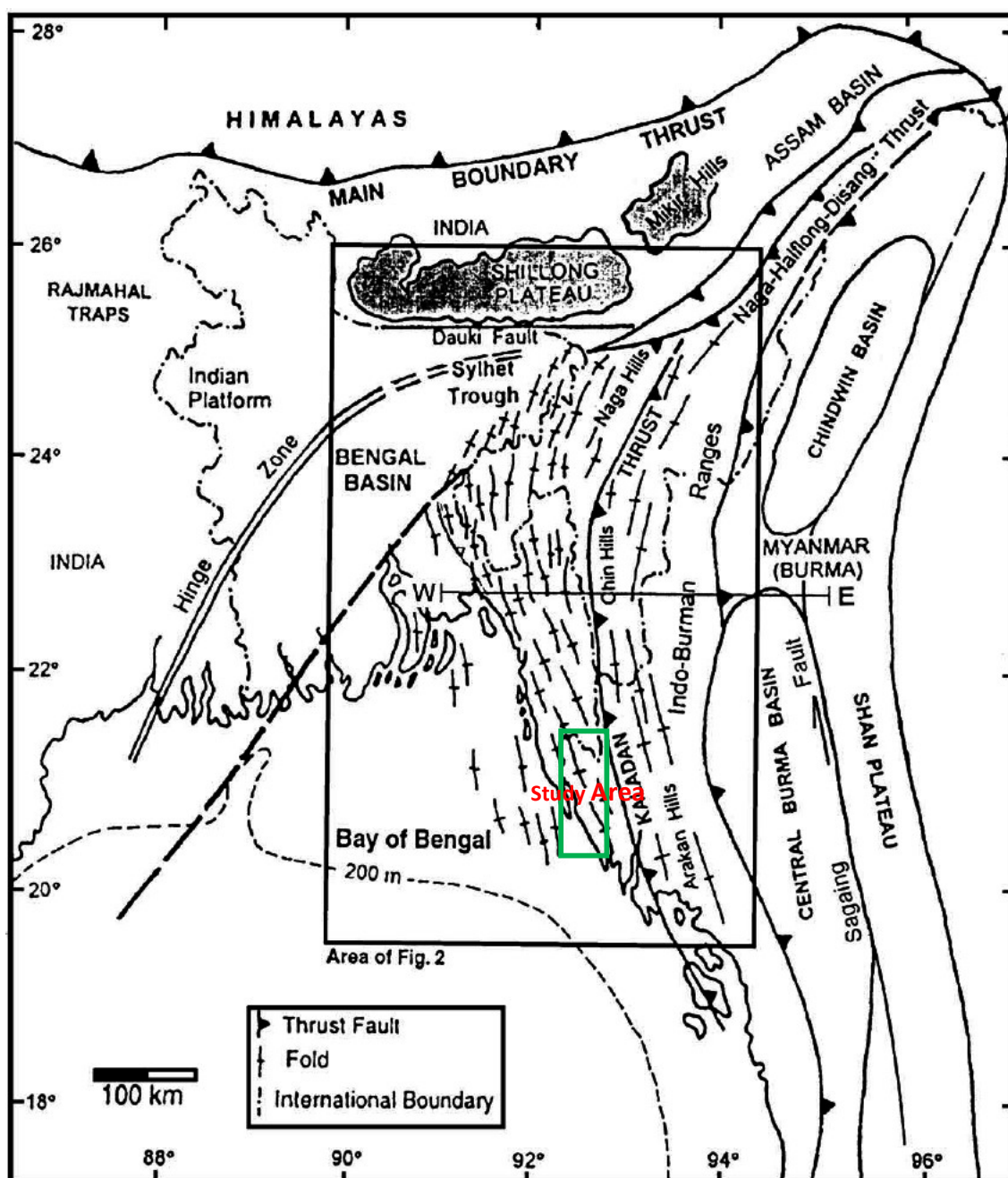


Figure 2.2: Regional tectonic setting of the Bengla Basin showing its tectonic elements. The hinge zone separates the shallow Indian Platform from the central deep basin and the Eastern fold belt (Alam, 2003).